

Where Comparison Rankings Enter Open-Team Reward Shaping

Research phase: DRAFT after 1 round, 0 ITERATE, 0 REVISE. Edit phase: paper accepted at review round 1

Abstract

Q asks that a prediction guarantee be connected to a downstream objective and that predictor calls be charged; P turns multimodal pairwise comparisons into agent-specific potential-based rewards. We analyse the interface between these claims, without asserting a new return guarantee. With complete per-agent reward accounting, P's shaping telescopes from exact discounted returns and cancels from a matched critic's temporal-difference residual. With full-population pooling, its softmax normalisation also removes every individual ranking from the pooled shaping increment. An active-only reward mask instead leaves a reentry term and can preserve ranking dependence at coalition changes. These are conditional consequences of the stated potential, not findings about P's implementation. A two-agent sample additionally shows that connectedness alone does not ensure the finite unregularised Bradley–Terry estimate required by P's stated error bound. The remaining question is whether comparisons improve finite-budget unshaped performance through a measured training channel after their calls are charged.

1 The two papers and the question

Q surveys learning-augmented algorithms and distinguishes predictor error from the downstream cost it is meant to control. Its inference-graph account requires a justified edge between components, and its prediction-pricing proposal explicitly counts calls in objective units [4]. P's MARS-RA queries a multimodal comparator for each ordered pair in the active coalition, fits Bradley–Terry scores, softmaxes them into a vector potential, and adds the potential difference to a scalar environment reward for multi-agent training [3]. It reports task-success and comparison-accuracy trends; its estimation proposition addresses latent comparator scores, while its Shapley proposition assumes alignment of those scores with Shapley values. Neither proposition states a bound on the return of the learned policy per comparison call.

We ask which reward or critic route could carry P's ranking signal into a finite-training improvement when Q's downstream and cost requirements are applied. The calculations below identify routes to inspect. They do not establish that any route is used by P's implementation.

2 Conditional accounting for P's potential

Let $\mathcal{N}$ be the fixed agent population, $I^t \subseteq \mathcal{N}$ the active coalition, and $a_{i,t} = \mathbf{1}\{i \in I^t\}$. For a nonempty nonterminal coalition, let $\psi_{i,t}$ be P's softmax score for $i \in I^t$, embedded as zero for an inactive agent. Use one cached value at each endpoint of adjacent transitions and set $\psi_{i,T} = 0$ at termination. Write $F_{i,t} = \gamma\psi_{i,t+1} - \psi_{i,t}$, with shaping weight ρ . This fixed-coordinate, cached-potential convention is a condition of our analysis; P does not specify its cross-coalition reward convention.

Proposition 1 (Complete accounting and active masking). *If agent i 's shaping increment is credited on every transition, then, pathwise,*

$$\sum_{t=0}^{T-1} \gamma^t F_{i,t} = -\psi_{i,0}.$$

If it is credited only when $a_{i,t} = 1$, then

$$\sum_{t=0}^{T-1} \gamma^t a_{i,t} F_{i,t} = -a_{i,0} \psi_{i,0} + \sum_{t=1}^{T-1} \gamma^t (a_{i,t-1} - a_{i,t}) \psi_{i,t}.$$

At a reentry time t , the latter sum includes $-\gamma^t \psi_{i,t}$.

Proof. In the first sum, every intermediate $\psi_{i,t}$ appears once with each sign, leaving $-\psi_{i,0} + \gamma^T \psi_{i,T}$. In the masked sum, the coefficient of $\psi_{i,t}$, for $1 \leq t < T$, is $\gamma^t (a_{i,t-1} - a_{i,t})$; the terminal term is zero. $\square$

The first identity is the standard potential-shaping cancellation applied to P's coordinates. For a fixed initial potential determined before the first action, its pathwise offset does not change exact policy ordering. The second identity shows why zeroing inactive potentials is insufficient when rewards are masked: the transition into a returning agent is not credited to that agent. Neither identity says how P actually routes its rewards.

Proposition 2 (Pooling fork). *Let $u_t = \sum_{i \in \mathcal{N}} \psi_{i,t}$. If every coordinate is pooled, the shaping increment is $\sum_{i \in \mathcal{N}} F_{i,t} = \gamma u_{t+1} - u_t$. P's softmax gives $u_t = 1$ for each nonempty nonterminal coalition, so this pooled increment is independent of the individual rankings. If instead only $i \in I^t$ are pooled and both endpoint coalitions are nonempty with a nonterminal successor, then*

$$\sum_{i \in I^t} F_{i,t} = \gamma \left(1 - \sum_{i \in I^{t+1} \setminus I^t} \psi_{i,t+1} \right) - 1.$$

Thus entrant scores can affect this active-only pool.

Proof. Sum the potential differences, then use $\sum_{i \in I^t} \psi_{i,t} = 1$. In the active-only case, the successor sum includes exactly the agents active at both endpoints, leaving out entrants. $\square$

The proposition concerns the shaping increment, rather than a claim that P uses a pooled team reward. Separate agent rewards can still change approximate training updates, even where complete per-agent returns telescope.

For a critic $\widehat{V}_i^{\text{env}}$, set $\widetilde{V}_i = \widehat{V}_i^{\text{env}} - \rho \psi_i$. Under complete accounting its shaped temporal-difference residual is exactly the unshaped residual. If the implemented critic adds mismatch e_i , the residual difference is $\gamma e_{i,t+1} - e_{i,t}$. The same cancellation holds for a generalised-advantage sum formed from matched residuals. This identifies critic mismatch as a possible training channel, not a bound on final return.

3 Two limits on transferring an accuracy claim

Proposition 3 (Connected comparisons need not yield a finite estimate). *For two agents at a true Bradley–Terry tie, take two independent comparisons of their sole pair. The undirected comparison graph is connected, but the unregularised maximum-likelihood estimate is not finite on the event that both outcomes favour the same agent. This event has probability $1/2$.*

Proof. Each ordered draw favours either agent with probability $1/2$. Each unanimous outcome has probability $1/4$. For either outcome, increasing the winning agent’s score difference strictly increases the likelihood, whose supremum is reached only at an infinite difference. $\square$

P states a finite score-error guarantee with probability at least $1 - n^{-2} = 3/4$ under connectedness. The example permits a finite estimate on at most half the samples, so connectedness by itself cannot support that assertion for its unregularised fit. This is an objection to that stated estimator premise, not to P’s reported experiments; it does not assess a constrained or regularised estimator.

A distinct one-step diagnostic also limits what pairwise accuracy can imply. If agent 2 is inert, agent 1 chooses $a \sim \text{Bernoulli}(1/2)$, $G = a$, and the logit score is $z = a - 1/2$, then the baseline estimator $g_b = z(G - b)$ has mean $1/4$ and variance $(b - 1/2)^2/4$. The neutral $b = 1/2$ has zero variance; a rank-correct softmax of Shapley scores $(1/2, 0)$ gives $b > 1/2$ and positive variance. This example tests a proposed downstream error measure, not MAPPO performance.

4 Prior work and interpretation

The central cancellation is published potential-shaping theory, rather than a new invariance theorem. P itself cites dynamic shaping work; Gupta and colleagues show that potential shaping can leave expected policy-gradient updates unchanged while increasing their variance [1]. Müller and Kudenko already obtain a shaping potential from vision-language pairwise preferences and study label accuracy and learning speed in single-agent environments [2]. Thus using visual preferences for potential shaping, and warning that ranking quality alone does not establish gradient quality, are existing steps.

What this note contributes for the Q–P pair is the explicit three-way accounting fork: complete per-agent rewards telescope, full-population pooling erases P’s softmax ranking, and active-only credit can expose entrant scores while leaving a reentry term. The fork is algebraic and conditional. It does not establish novelty over all reward-routing implementations, nor does it identify P’s implementation.

P’s stated all-ordered-pairs protocol calls the comparator $q_t = |I^t|(|I^t| - 1)$ times at a queried step. P reports 50-million-step training times of 16 hours for MAPPO and 30 hours for MARS-RA in its setting. These observations motivate measuring held-out, unshaped success or return under fixed environment-step and wall-clock budgets, with comparison count and latency reported separately. They do not yield a cost-adjusted performance bound.

5 Limitations and open questions

P’s displayed method leaves the inactive-agent reward mask, cross-coalition embedding, pooling rule, and critic targets unresolved, so the conditional identities cannot diagnose its realised updates. Repeat comparator calls may be correlated, and the toy variance calculation is a frozen-policy diagnostic. We have no controlled finite-training experiment, no measured comparison-to-return sensitivity, and no theorem of improved learned-policy return.

A direct audit would trace a coalition entry or reentry through P’s reward tensor, inactive-agent mask, and critic target. Matched-critic, neutral-potential, and scale-matched random-potential controls could then compare held-out unshaped outcomes at fixed step and time budgets while counting calls. Whether a ranking effect survives, and whether it pays for those calls, remains open.

References

- [1] Dhawal Gupta, Yash Chandak, Scott M. Jordan, Philip S. Thomas, and Bruno Castro da Silva. Behavior alignment via reward function optimization, 2023. URL: <https://arxiv.org/abs/2310.19007>, [arXiv:2310.19007](#).
- [2] Henrik Müller and Daniel Kudenko. Automating potential-based reward shaping with vision language model guidance, 2026. URL: <https://arxiv.org/abs/2606.27180>, [arXiv:2606.27180](#).
- [3] Dawei Wang, Di Zhao, Xinyuan Liu, Marci Chi Ma, Xiaoyang Liu, Chengming Zhou, Gary Ushaw, and Richard Davison. MARS-RA: Rank aggregation for credit assignment via multimodal comparisons in embodied multi-agent cooperation, 2026. URL: <https://arxiv.org/abs/2607.27967>, [arXiv:2607.27967](#).
- [4] Hailiang Zhao, Xueyan Tang, Peng Chen, Zenong Ye, Tianjun Zhu, Jianwei Yin, and Shuiguang Deng. Learning-augmented algorithms: Guarantees, construction mechanisms, and system-level implications, 2026. URL: <https://arxiv.org/abs/2609.04787>, [arXiv:2609.04787](#).